Aditya Kumar

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Education

Dayananda Sagar College of Engineering — **Bangalore, India**
B.E. in Computer Science and Engineering — *2025 – 2029*

Technical Skills

Languages: Go, JavaScript, TypeScript, Python
Databases: PostgreSQL, MongoDB, Redis, Firebase, Supabase
Frameworks: Next.js, Gin, React, Node.js, Express
Developer Tools: Git, Docker, Kubernetes, Postman, Linux
Libraries: Socket.IO, React Map GL, WebRTC, WebSockets, Cobra, Bubble Tea

Open Source Contributions

- Surge **Contributed 4+ pull requests** across the **daemon's REST layer**, **download management**, and **TUI**; expanded the API to core-service parity with 9 new endpoints, support for custom download IDs, and fixed TUI issues.
- Greenmask **Contributed** to an open-source **PostgreSQL data anonymization tool** by fixing a **non deterministic masking bug** in **Go** caused by randomized map iteration order, ensuring **reproducible output** across runs.

Projects

Cadence | *Go, Next.js, PostgreSQL, Redis, FFmpeg, OpenRouter*
- Built an **AI-powered video clipping platform** that automatically extracts **shareable, engaging clips** from long-form YouTube content like podcasts and streams, using a **Go** backend and **Next.js** frontend.
- Designed an **async job pipeline** with **Redis** as a message queue: downloads videos via **yt-dlp**, transcribes subtitles, batches transcripts to an LLM for **scene detection**, and cuts clips with **FFmpeg**.
- Engineered a **multi batch LLM prompting** strategy with **retry logic and exponential backoff**, scoring clips on **hook strength and shareability**, then ranking to a **dynamic target count** based on video duration.

ArmoraCrypt | *Go, Cobra, Dropbox API*
- Built a **CLI tool** for local **AES-256 encryption/decryption** of files and folders, enabling secure uploads and downloads to cloud storage while safeguarding against **provider surveillance** and unauthorized access.
- Integrated **zipping and unzipping** to handle entire folders, streamlining encryption for complex directory structures.
- Designed a **master-key initialization system** for generating and managing local encryption keys, ensuring a secure, self-contained **key management** system.
- Implemented **upload and download** of encrypted files via the **Dropbox API**, allowing users to safely store and retrieve sensitive data.

Droppa | *WebRTC, Socket.io, Next.js*
- Built a **P2P file transfer app** using **WebRTC DataChannels** for direct device-to-device file transfers.
- Implemented **chunked transfers** with **backpressure handling** to control memory usage when transferring **multi-GB files** over unreliable connections.
- Built a **WebSocket signaling server** for **SDP offer/answer** and **ICE candidate exchange**, with dynamic **QR codes** for easy session setup.

BlackBox | *Go, Bubble Tea (TUI)*
- Built a high-speed, concurrent **HTTP download manager** in **Go**, using **goroutines** and **mutex** based synchronization to queue and process multiple file downloads in parallel without data races.
- Designed an interactive **Terminal UI** using **Bubble Tea** and **Lip Gloss**, featuring real-time progress bars, live download speeds, and ETA tracking for each queued download.
- Improved download speed using **chunked parallel fetching**, with the downloader running on a **separate goroutine** so the UI stays responsive during large transfers.

Achievements

One of 40 participants selected for the **ACM Summer School 2026** at **IIT Hyderabad** on Symmetric-Key Cryptography.

Runner-up at the **Beyond Abstraction Hackathon** (**ETHIndia**), awarded **$1,000**.

Volunteering

Point Blank Member — **Bengaluru, India**
Multidisciplinary programming community ***(IndiaFOSS Student Community of the Year)*** — *Dec 2025 – Present*